

Indian Institute of Technology Indore

Science & Technology Council
IITISoC 2026 — Project Proposal

Selected Problem Statement (Preference 1):

Personalized Mental Health Digital Twin using AI

Domain: Artificial Intelligence & Machine Learning *Difficulty:* Advanced *Team Size:* 4–6 Members

1. Team Details

1.1 Team Leader

Name Rohith M S
Roll Number 250005038
GitHub <https://github.com/rohithsuresh21>
LinkedIn <https://www.linkedin.com/in/rohith-m-s-964511378>
Skills PyTorch, Neural Networks, RAG, Transformers, LLMs, Feature Engineering, Time-Series Analysis

1.2 Other Team Members

#	Name	Roll Number	Key Skills
2	—	—	—
3	—	—	—
4	—	—	—

2. Problem Statement Preferences

Preference	Problem Statement	Domain
1	Personalized Mental Health Digital Twin using AI	AI / ML
2	Building Lightweight Multi-Task AI with MALORA	AI / ML
3	Pathway-Powered Agentic Gateway	AI / ML

3. Problem Understanding

Mental health conditions develop gradually and vary significantly across individuals, making early detection and timely intervention a major challenge. Traditional AI systems rely on static binary labels (e.g., *depressed* / *not depressed*) that fail to capture the continuous, deeply personal nature of human cognition and behavior.

This project aims to develop a **Personalized Mental Health Digital Twin (PMHDT)** — an AI-driven virtual representation of an individual’s mental state that:

- Continuously learns from the user’s behavioral and linguistic data.
- Establishes a personalized baseline rather than relying on population-level norms.
- Detects subtle deviations from that baseline as early warning signals.
- Predicts future mental health risks using longitudinal data.
- Delivers interpretable, trust-building explanations for users and clinicians.

4. Proposed Solution

Our solution is structured around five tightly integrated modules that together constitute the digital twin pipeline.

4.1 Module 1 — Personalized Baseline Behavior Modeling

We will learn individual-specific behavioral fingerprints from multimodal inputs:

- **Textual signals:** User journals, chat logs, and free-form writing — encoded via a fine-tuned transformer (e.g., MentalBERT or a quantized LLaMA variant) to capture linguistic complexity, vocabulary richness, and semantic coherence.
- **Interaction signals:** Typing dynamics (keystroke timings, edit rates), session duration, and app-usage patterns captured through a lightweight client SDK.
- **Optional signals:** Speech prosody features (speaking rate, pitch variation) and wearable data (HRV, sleep cycles) if available.
- Baseline profiles will be updated incrementally using an exponential moving average combined with an online anomaly threshold recalibrator.

4.2 Module 2 — Temporal Mental State Modeling

Rather than discrete labels, we model mental state as a **continuous trajectory** evolving over time:

- A **Temporal Fusion Transformer (TFT)** or lightweight **LSTM-based encoder** will embed longitudinal feature sequences into a latent mental-state space.
- The latent representation tracks gradual drift (slow mood decline) as well as sudden anomalies (acute stress spikes).
- Sliding-window context windows (e.g., 7-day and 30-day) will capture both short-term fluctuations and longer-term trend shifts.

4.3 Module 3 — Deviation & Anomaly Detection

Personalized thresholds (not population norms) will be used to flag early warning signals:

- **Linguistic anomalies:** Drops in sentence coherence, increased repetition, reduced vocabulary diversity.
- **Behavioral anomalies:** Irregular session timings, sudden drop in engagement, erratic typing patterns.

- Detection methods: **Isolation Forest** and **LSTM-Autoencoder reconstruction error** with user-specific adaptive thresholds.
- Confidence scores will accompany every alert to avoid over-triggering.

4.4 Module 4 — Predictive Risk Modeling

Proactive forecasting of upcoming risk states (7-day horizon):

- **Target risks:** Stress escalation, burnout onset, depressive tendency, social withdrawal.
- **Model:** Multi-output probabilistic forecaster (TFT or N-BEATS with uncertainty quantification via Monte Carlo Dropout).
- Outputs are risk probability distributions, not binary flags, giving users and clinicians actionable confidence levels.

4.5 Module 5 — Explainability & Feedback Dashboard

Trust and transparency are core requirements:

- **SHAP values** on text features will highlight which linguistic markers drove a risk flag.
- **Attention-weight visualizations** from the temporal model will show which time points were most influential.
- Natural-language summaries (generated by a grounded LLM module) will translate numeric signals into plain English, e.g., *"Significant drop in writing coherence detected over the past 5 days."*
- A **clinician-facing view** and a **user-facing view** will be provided with appropriate levels of detail.

5. System Architecture Overview

High-Level Data Flow:

User Inputs (text / interaction / optional speech)

↓ *Feature Extraction & Preprocessing*

Baseline Profiler → Temporal Encoder (TFT / LSTM)

↓ *Latent Mental-State Vector*

Anomaly Detector || Risk Forecaster

↓

Explainability Engine (SHAP + Attention) → Dashboard (User / Clinician)

Technology Stack:

Component	Tools / Libraries
Language & Frameworks	Python 3.10+, PyTorch, HuggingFace Transformers
Temporal Modeling	Temporal Fusion Transformer, LSTM-Autoencoder
NLP / Text Features	MentalBERT / sentence-transformers, spaCy
Anomaly Detection	scikit-learn (Isolation Forest), PyOD
Explainability	SHAP, Captum
Backend API	FastAPI
Frontend / Dashboard	Streamlit or React + Recharts
Data Storage	SQLite (local) / PostgreSQL (server)
Version Control	Git + GitHub

6. Ethical Considerations & Privacy

- **Data minimization:** Only signals necessary for baseline modeling are collected; raw text is never stored beyond the session unless explicitly consented to.
- **Local-first inference:** All model inference runs locally on the user’s device by default, ensuring no sensitive data leaves the device.
- **Consent & control:** Users can view, export, or delete their entire behavioral profile at any time.
- **Not a clinical tool:** The system will prominently communicate that it is a monitoring aid and not a substitute for professional mental health care.
- **Bias auditing:** Baseline models will be evaluated across demographic subgroups to detect and mitigate systematic bias in risk scoring.

7. Evaluation Plan

Component	Metric	Target
Anomaly Detection	Precision / Recall on held-out simulation set	>0.80 / >0.75
Risk Forecasting	MAE, CRPS (probabilistic)	Competitive with LSTM baseline
Explainability	SHAP faithfulness check (perturbation test)	Pass
User Dashboard	Task completion & qualitative usability test	Satisfactory
Privacy	Zero external data transmission audit	Pass

8. Project Timeline

Phase	Duration	Deliverables
Phase 1	Week 1–2	Literature review, dataset selection, environment setup, baseline feature extraction pipeline
Phase 2	Week 3–4	Baseline profiler + temporal encoder implementation and initial training
Phase 3	Week 5–6	Anomaly detection and risk forecasting modules, initial integration
Phase 4	Week 7–8	Explainability layer, dashboard (user & clinician views), end-to-end integration
Phase 5	Week 9–10	Evaluation, ablation studies, ethical audit, bug fixes and refinement
Phase 6	Week 11	Final report, demo video, GitHub cleanup and submission

9. Expected Deliverables

1. **Functional Digital Twin Prototype** — end-to-end system running locally.
2. **Personalized Baseline Modeling Module** — with online update capability.
3. **Temporal Deviation Detection System** — with user-specific thresholds.
4. **Risk Prediction Model** — with probabilistic outputs and evaluation metrics.
5. **Explainability Dashboard** — dual view for users and clinicians.
6. **Clean GitHub Repository** — reproducible, well-documented codebase.
7. **Final Technical Report** — research-paper style, covering methodology, results, limitations, and ethical considerations.

This proposal is submitted for IITISoC 2026 under the Science & Technology Council, IIT Indore.